

Development of Exergame to Resolve Deconditioning in Children with Orthostatic Dysregulation

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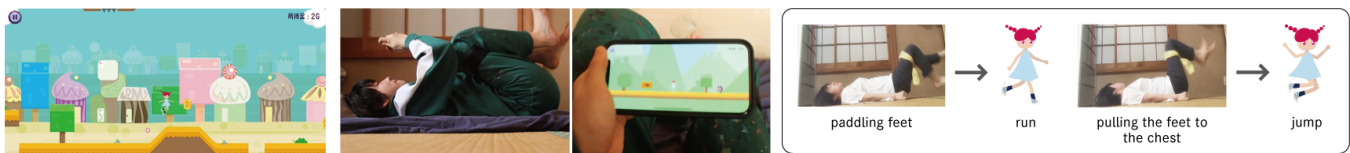


Figure 1: Our exergame. (Left) Game screen. (Middle) Play on smartphone in lying posture. (Right) Correspondence between motion and movement.

ABSTRACT

In this study, we proposed an exergame to reduce resistance to exercise and maintain motivation in adolescents with orthostatic dysregulation. We created a 2D side-scrolling action game in Unity to be played on smartphones. The game is synchronized with exercises that can be done in a lying posture. We analyzed the effect of the game on the feelings of the participants. Our experiments showed that the use of the exergame has a positive effect on the participants' emotions during exercise.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction(HCI).**

KEYWORDS

exergame, gamification, orthostatic dysregulation

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1 INTRODUCTION

In recent years, serious and prolonged cases of orthostatic dysregulation (OD), an autonomic nervous system disorder that causes dizziness and palpitations upon standing up, have been increasing, and the number of adolescents with serious conditions resulting in school absence has been increasing. One of the reasons for this is deconditioning, i.e., declining physical function due to long periods of inactivity. Therefore, exercise therapy is considered an effective treatment method.

Ishizaki showed in a bed rest experiment on students that healthy young people who do not move their bodies at all for 10 days fall into a state of deconditioning that leads to orthostatic intolerance (the adult version of orthostatic dysregulation)[Ishizaki et al. 2004]. Given this result, children with OD who are in bed at home require rehabilitation to prevent serious cases of the disorder. Regarding exercise therapy for orthostatic dysregulation, Yanagimoto [Yanagimoto 2021] introduces leg exercises in the lying, semi-lying, or sitting position. Although these therapies have been shown to be effective, Yanagimoto notes that there are challenges in terms of maintaining motivation for patients to exercise.

Khan et al. have shown that exergames are effective in stroke rehabilitation in terms of motivation and enjoyment[Khan et al. 2018]. Therefore, the use of exergames can be considered effective in increasing motivation for rehabilitation and treatment.

Based on the above, this study develops an exergame to maintain motivation for exercise in children with OD. The game uses leg movements (bending and paddling) that can be done in bed as input to accommodate patients with difficulties exercising while standing due to OD symptoms. Our experiment investigates the effect of the exergame on emotional state after exercise.

2 DESIGN AND IMPLEMENTATION

We created a 2D side-scrolling action game to be played on smartphones. The game was created in Unity and is synchronized with exercises that can be done in a lying posture. OPRHE CORE and OPRHE TRACK Hub (both made by OPRHE Inc.) are used to capture exercise. OPRHE CORE is a device that is equipped with a 6-axis motion sensor (3-axis accelerometer and 3-axis gyroscope), which can calculate quaternion and Euler angle from acquired 6-axis data. OPRHE TRACK Hub is a Mac OS software that receives data captured and calculated by OPRHE CORE via Bluetooth connection and enables application development using this data. Data captured by OPRHE CORE is sent to Unity via OPRHE TRACK Hub and projected to the game.

2.1 System design

The system diagram is shown in Figure 2. The OPRHE CORE attached to the user's body sends values such as angle and acceleration to the OPRHE TRACK Hub in real-time. The OPRHE CORE is connected to the computer via Bluetooth. By using the OPRHE TRACK Hub and Unity OSC, we implemented OSC (Open Sound Control) communication between the OPRHE TRACK Hub and Unity. Through OSC communication, the values received by OPRHE TRACK Hub are sent to Unity and then reflected in the game.

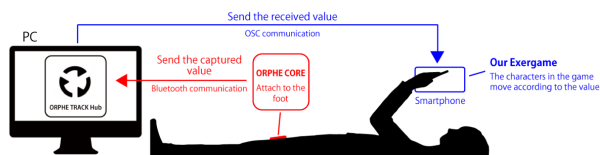


Figure 2: System diagram

2.2 Game design

The rules of the game are simple: the player runs to the goal while avoiding enemy characters and holes by jumping. Along the way, the player aim for a high score by picking up coins and defeating enemies. The game graphics considered the user target age (middle and high school students). In addition, the game was designed to be 2D (Figure 1) to avoid the possibility of video sickness.

2.3 Motions used in the game

The two motions used for input are "paddling feet in a lying posture" and "pulling the feet to the chest in a lying posture." Each motion is linked to the movement of the characters in the game, as shown in Figure 1. The paddling motion was detected by the acceleration of the OPRHE CORE, the pulling motion was detected by the angle. Since the angle during the pulling motion depends on the user's physical shape, this game is implemented a calibration function.

3 EXPERIMENT

An experiment was conducted to verify the effectiveness of the implemented game in maintaining motivation. The participants were one male and five female healthy university students. Participants were required to "paddle their legs in a lying posture" with and without the game. Positive and negative levels before and after

each exercise were measured using Clark's J-PANAS method[Clark 1989], and the emotion changes compared.

3.1 Experiment conditions

The experimental flow is as follows. Participants were divided into Pattern 1 and Pattern 2 for the experiment to eliminate order effects.

Pattern 1 participants first exercised with the game, took a 15-minute break, and then exercised without the game. Before and after each exercise, they answered a survey about their emotions using the J-PANAS method. In Pattern 2, exercise without the game was performed first. In both exercises with and without the game, 30 foot paddling exercises were performed. In the case of exercise with the game, slightly more exercise occurred due to the input of the jumping motion (pulling the feet to the chest in a lying posture).

3.2 Result

The results of the experiment are shown in Figure 3. Positive emotion levels tended to increase for exercise with the exergame, while tending to decrease for exercise without the game. The increase in negative emotion levels was slightly lowered for exercise with the game.



Figure 3: Experiment results(Change in emotion levels).

4 DISCUSSION AND CONCLUSION

Our experiments showed that exergame use has a positive effect on exercise. However, in practice it is necessary to design a game that people will want to play continuously. Future work will be needed to improve the implementation of the parts required for continuous play. For instance, we are considering creating scenarios with several stages in the game. We are also planning on ways of playing the game with only a smart phone and sensors, without via PC, in order to make it easier for people to play the game. In addition, as the current gesture recognition accuracy is not sufficient for recognizing many movements, improving this accuracy is another important task for future study.

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